Explainable AI-Based Financial Fraud Detection System

1. Introduction

Credit card fraud causes financial losses to customers and financial institutions.

This project uses machine learning to identify potentially fraudulent credit card transactions.

2. Problem Statement

Manual identification of fraudulent transactions can be difficult because transaction datasets may contain a very small number of fraud cases. Machine learning can help identify suspicious transaction patterns.

3. Objectives

• Understand and analyse credit card transaction data.

• Clean and preprocess the dataset.

• Explore transaction patterns using graphs.

• Train machine learning models for fraud detection.

• Save the trained models for future predictions.

• Demonstrate a transaction prediction using the trained model.

4. Dataset

The project uses the Kaggle Credit Card Fraud Detection dataset.

The dataset contains transaction features, transaction amount, and a class label.

The class label distinguishes normal transactions from fraudulent transactions.

5. Tools and Technologies

• Python

• Jupyter Notebook

• Pandas and NumPy

• Matplotlib and Seaborn

• Scikit-learn

• Pickle for saving trained model files

6. Methodology

1. Load the credit card transaction dataset.

2. Inspect the dataset and check its structure.

3. Perform data preprocessing and exploratory data analysis.

4. Visualise transaction patterns.

5. Train Logistic Regression and Random Forest models.

6. Save the trained models and feature scaler.

7. Load the saved files and demonstrate a prediction.

7. Machine Learning Models

Logistic Regression and Random Forest were used for classification.
Their performance should be evaluated using suitable measures such as precision, recall, F1-score, and confusion matrix. Fraud detection requires special attention to recall and class imbalance.

8. Explainability

Feature importance from the trained Random Forest model can help indicate which features influenced model predictions. Feature importance shows associations within the model and does not, by itself, prove causation.

9. Demonstration

A sample transaction was passed through the saved model and the output indicated a normal transaction. This is a demonstration only and does not establish overall model performance.

10. Limitations

• The model's predictions depend on the dataset and preprocessing.

• Fraud datasets are often highly imbalanced.

• A single sample prediction cannot establish model accuracy.

• Further validation and monitoring are needed before real-world use.

11. Future Scope

• Evaluate models using precision, recall, F1-score, and a confusion matrix.

• Add explainability tools such as SHAP, if available.

• Test the model on unseen transactions.

• Develop a simple user interface for transaction prediction.

12. Conclusion

This project demonstrates a machine-learning workflow for credit card fraud detection, including data analysis, preprocessing, model training, visualisation, and a sample prediction. Further evaluation is necessary before practical deployment.

13. Project Files

The project folder contains notebooks, trained model files, visualisations, and project documentation.